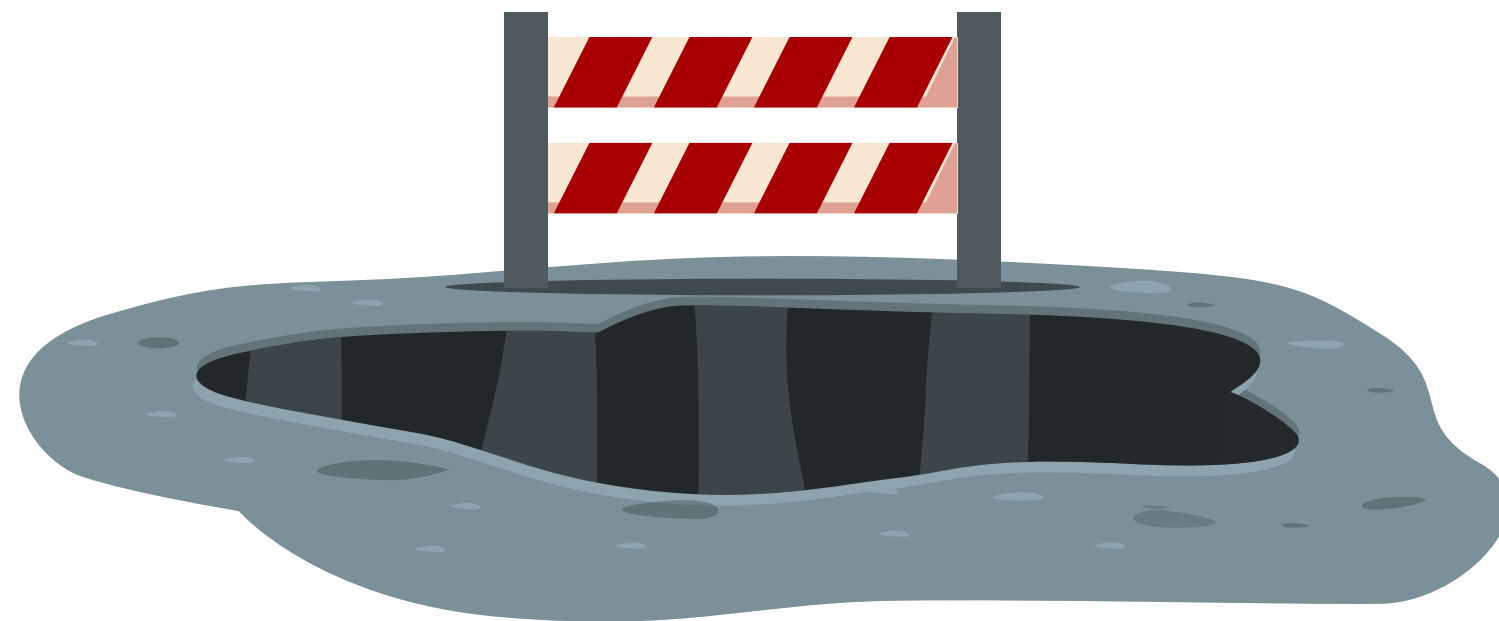


POTHOLE DETECTION

Unsupervised Pothole Segmentation using K-Means Clustering and Morphological Refinement.

Dataset : ARA 7.0 Road Surface Dataset

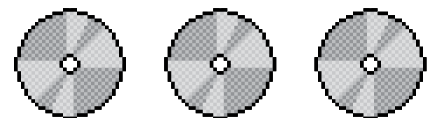
Stack : OpenCV + scikit-learn + scikit-image

Presented by Us



PROBLEM DEFINITION & MOTIVATION

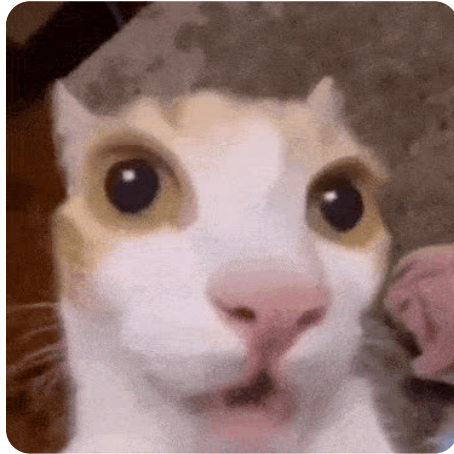
Road potholes are a leading cause of vehicle damage, accidents, and economic loss. Manual inspection is expensive and inconsistent. Automated detection from monocular RGB cameras offers a scalable, low-cost alternative.





AS MANY AS 27 TRAFFIC ACCIDENTS DUE TO POTHOLES OCCURRED THROUGHOUT JANUARY 2026.

As reported by the Directorate of Traffic (Ditlantas) of the Metro Jaya Police, 27 traffic accidents caused by potholes were recorded in the Jakarta metropolitan area from January 1 to January 28, 2026. This data highlights that within less than one month, dozens of crashes occurred solely due to road damage — resulting in at least one fatality, eight severe injuries, and twenty minor injuries.



THE BIGGER PICTURE OF INDONESIA'S ROAD CRISIS

01

**Regional Roads Are
Crumbling**

140,000 - 160,000 km of regional
roads are damaged
(30–40% of total)

02

Inefficient Manual Audits

Traditional road inspection relies
on slow, manual reporting, leaving
hazardous potholes undetected
and unrepaired for weeks.

03

**Annual traffic congestion
losses**

Traffic congestion alone costs
Indonesia Rp71 trillion annually

DATASET DESCRIPTION

ARA 7.0 Road Surface

Training Set

Consist of 450 road images with ground truth masks that mark the potholes as pixels

Test Set

Consists of 50 road images to test the model's generalization capabilities upon faced with unseen data

Data Characteristic

The images are captured from a high-angle, with various asphalt textures and visual noises such as tree shadows and water reflections

Only data with Mask can be evaluated for Metrics

IMAGE PREPROCESSING

Seq	Image Preprocessing	Function
1	BGR image + LAB Color Space	Select the L (Lightness) channel for lighting
2	Gaussian Blur	Local Lighting Estimate
3	Lightness/ (blur + ϵ) x 128	Shadow Normalization
4	CLAHE (Contrast Limited Adaptive Histogram Equalization)	Enhancing Local Contrast
5	Bilateral Filter	Smooth Out Noise and Preserve Edges

ROAD SEGMENTATION



Using Sky Mask (Hue 85-140) + Vegetation Mask (Hue 20-85) + Green Channel

OUR METHOD POTHOLE DETECTION

Model	Type	Method
Baseline	Unsupervised	Adaptive Tresholding
Advanced	Unsupervised	K-Means Pixel Clustering
Experiment	Supervised	Random Forest + SLIC Superpixel (9 fitur)

BASELINE: ADAPTIVE THRESHOLDING

Local threshold based on the intensity of surrounding pixels.

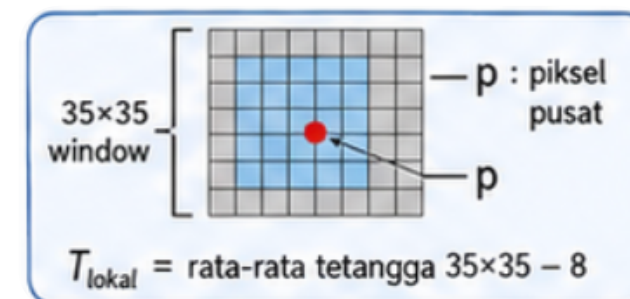
Input BGR



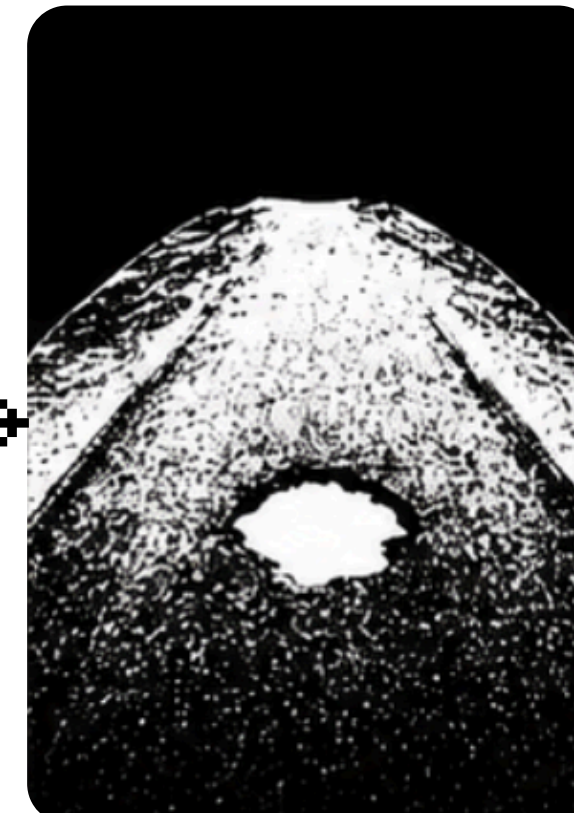
Grayscale



Adaptive Threshold



Inverse



Output Pred

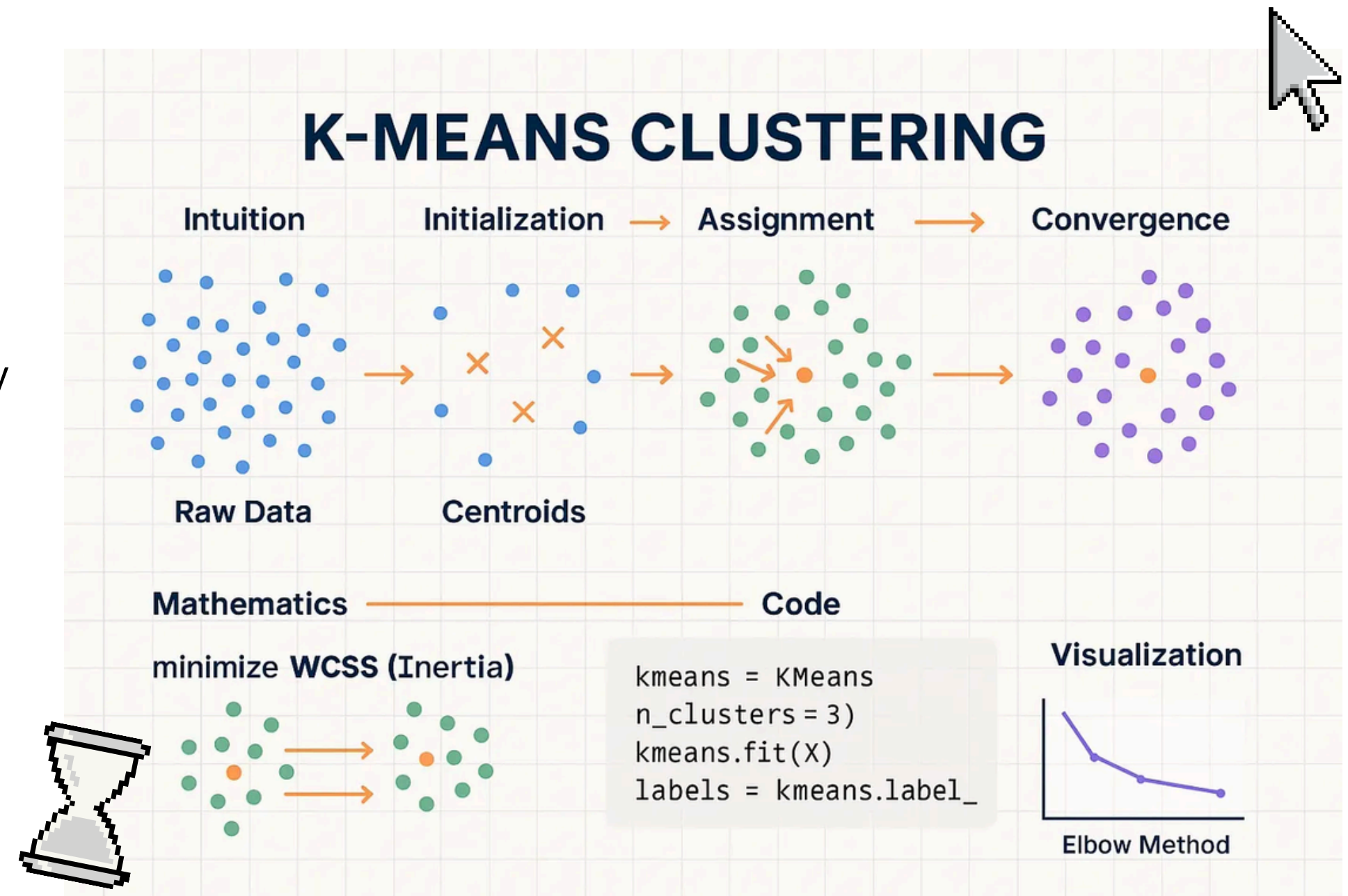
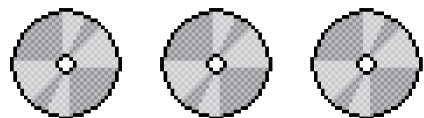


EARLY PLANS: K-MEANS CLUSTERING

Unsupervised Approach

Using K-Means to group pixels based on color intensity and textures automatically

Method: Finding an optimal centroid to separate normal asphalt and potholes



WHY DOES K-MEANS FAIL?

In initial testing, the unsupervised clustering method encountered several challenges:

Variation in Lighting

K-Means is sensitive towards lighting change. The same areas can have different clusters due to cloudy weather.

Shadow Ambiguity

Shadow of objects on the road can have pixels that are identical to a pothole, causing **False Positives** in the model.

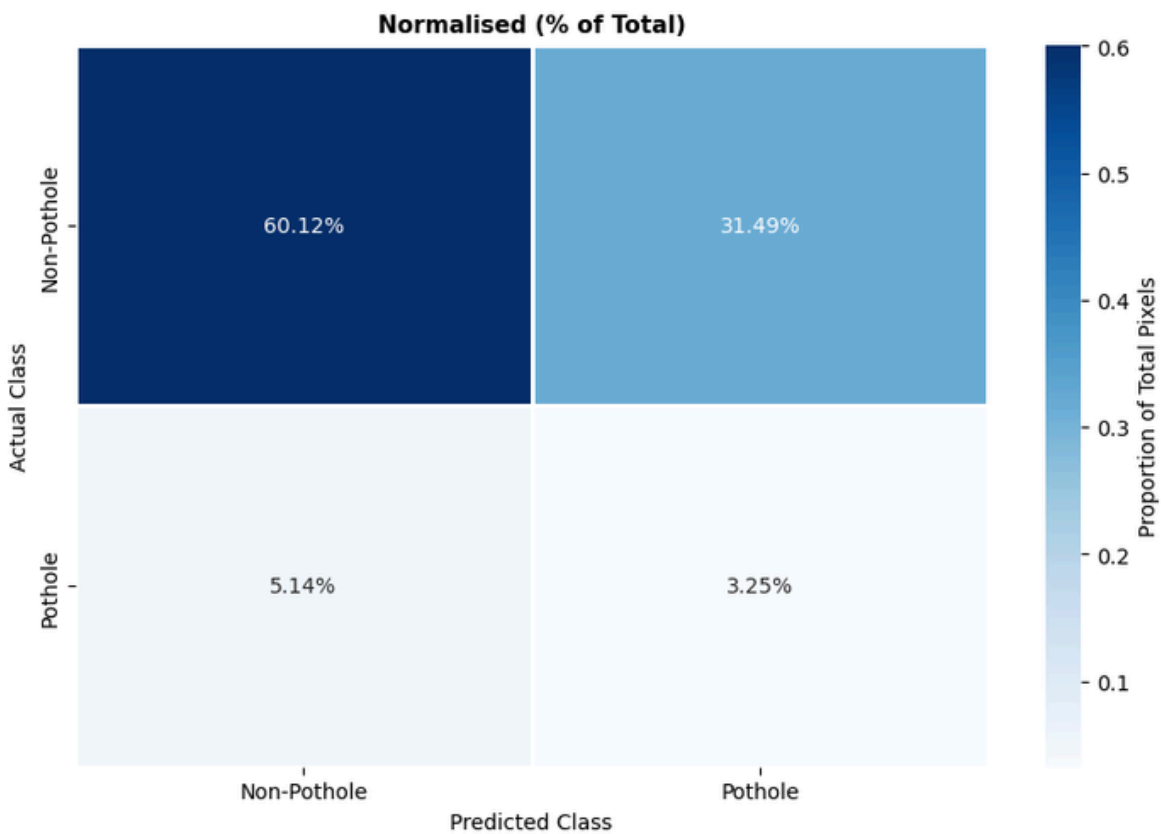
K parameter

Deciding on the correct and precise number of clusters (K) for every image is difficult and unstable due to varying roads in the image.

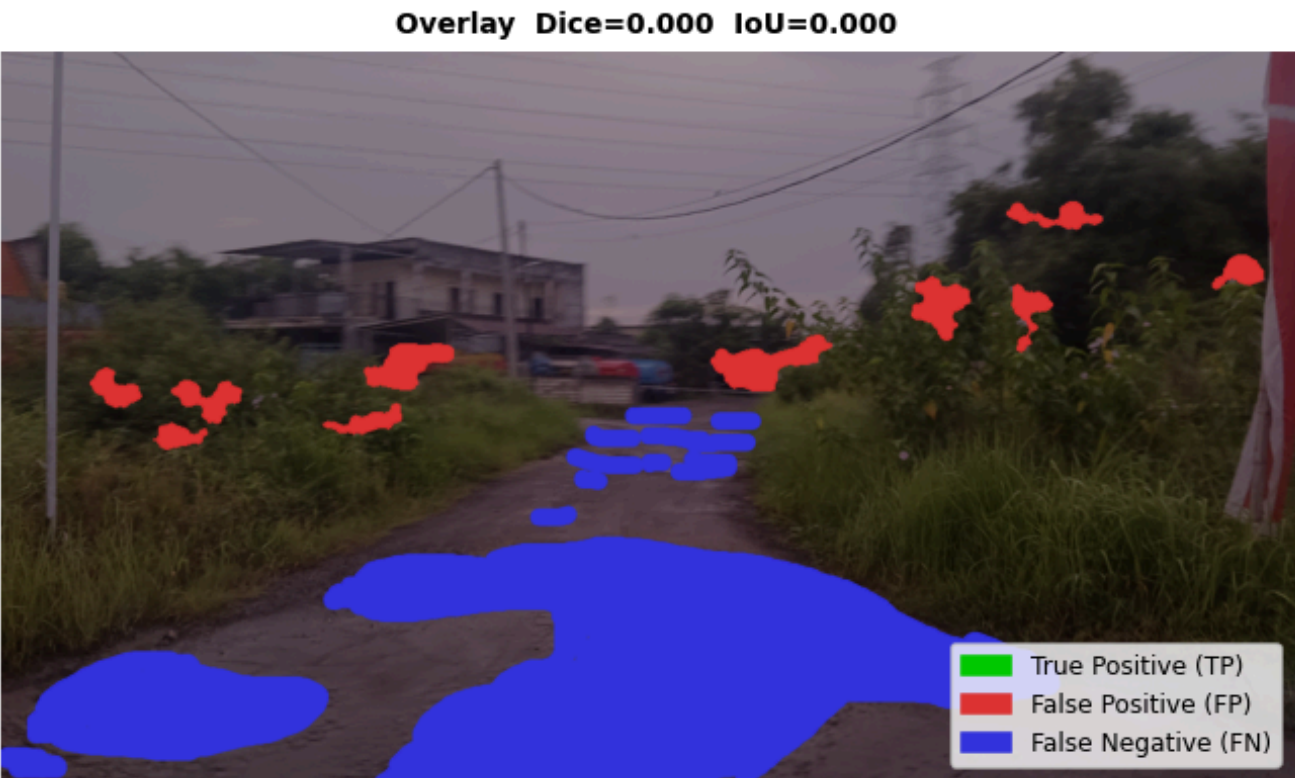




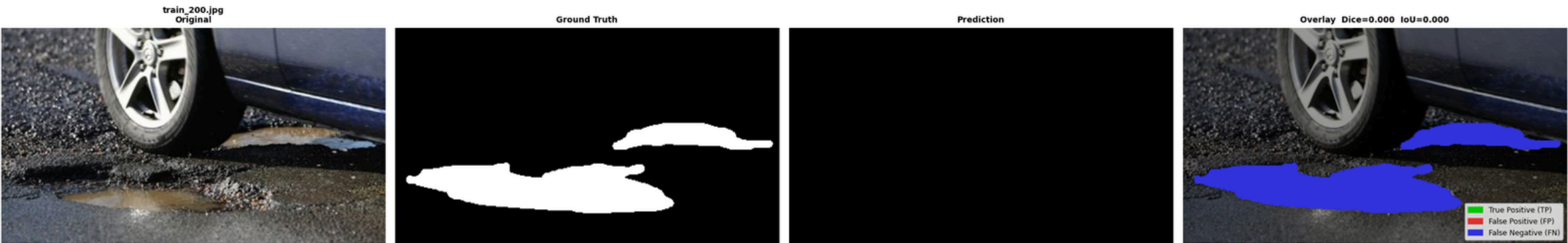
FAILURE EXAMPLE



Confusion Matrix



Segmentation Analysis: train_200.jpg



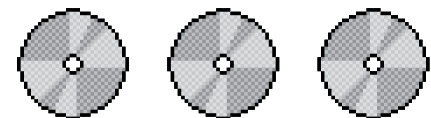


METHOD TRANSITION TO SUPERVISED ML

We decided to switch from unsupervised clustering to supervised classification.

Solution: Random Forest on Superpixels

- Using ground truth labels to “train” the model.
- Operates at the **region (superpixel)** level, rather than on individual pixels, making it faster and more consistent.
- Combines 9 different features (color, texture, edges).



File: train_464.jpg
Pixel IoU: 92.20%

Ground Truth Mask (Hijau)



Predicted Mask (Merah)



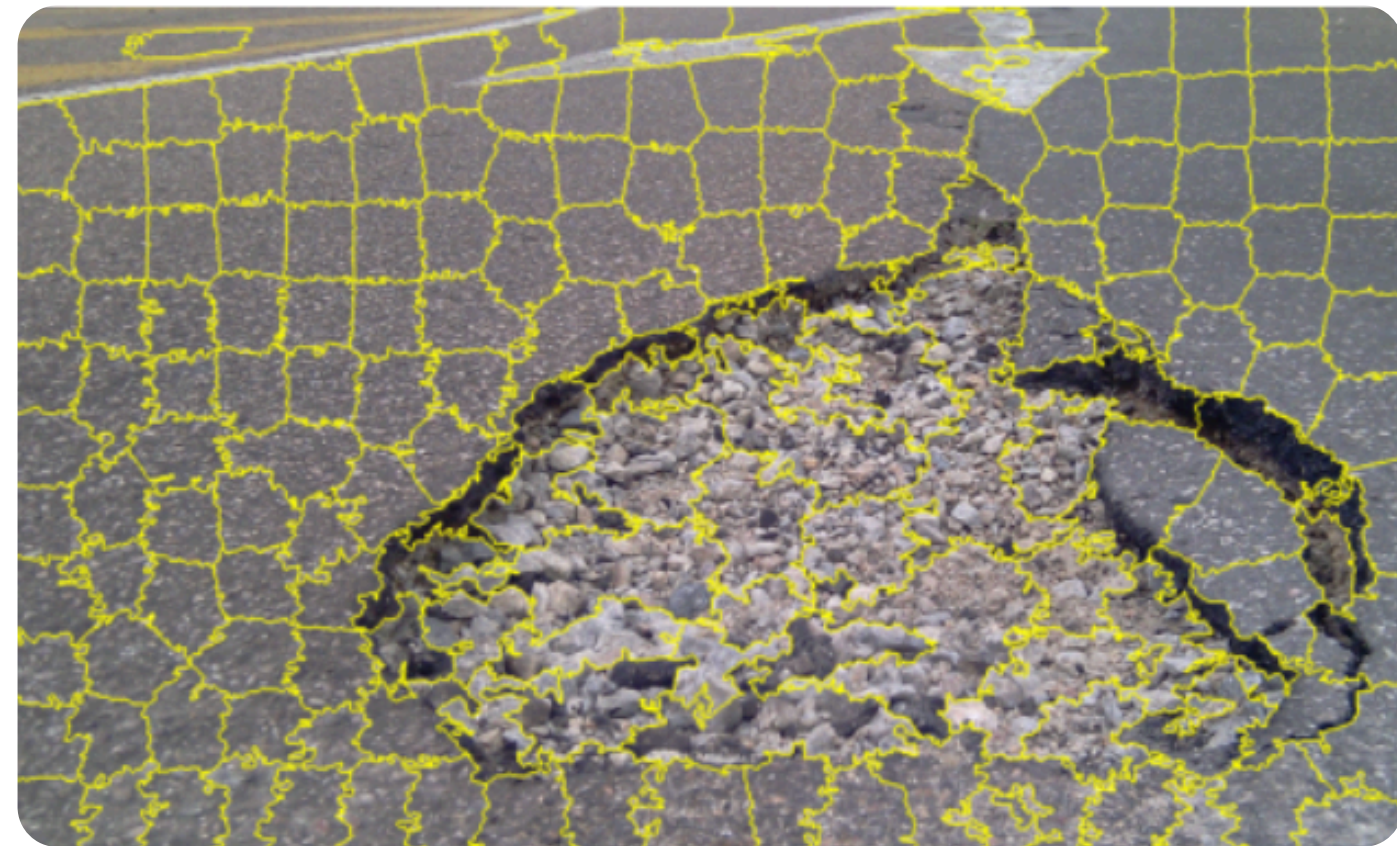
Example of successful detection.



RANDOM FOREST + SLIC SUPERPIXEL (9 FITUR)



Original Image



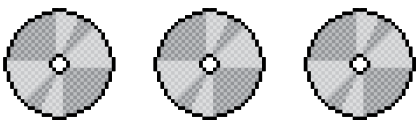
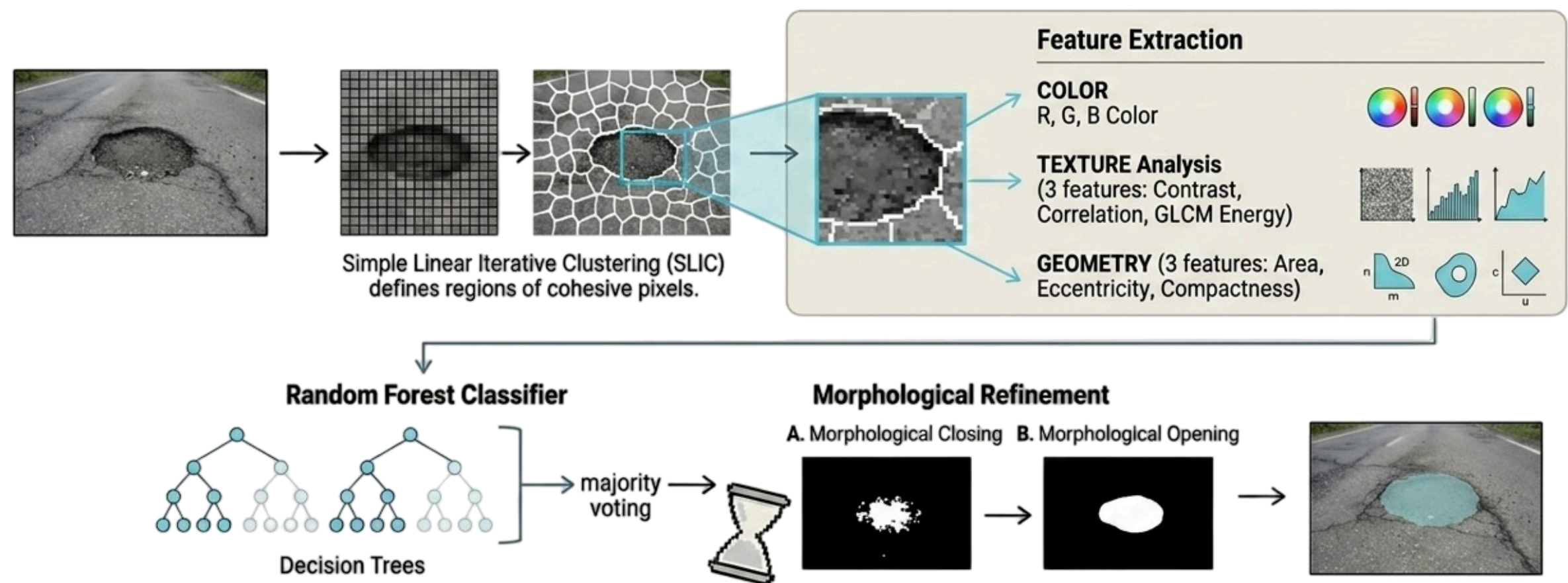
SLIC Super Pixel

EXPERIMENT: RANDOM FOREST + SLIC SUPERPIXEL (9 FITUR)

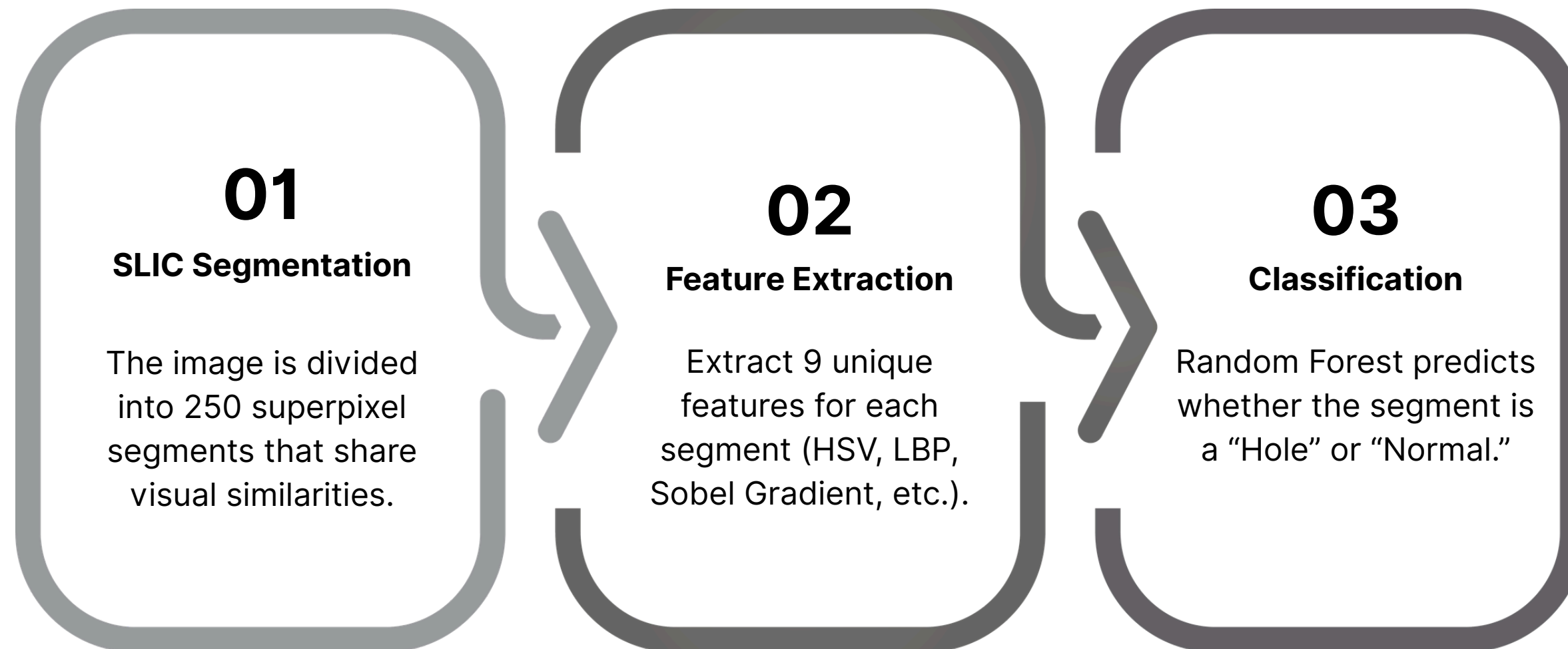
Supervised Approach

Grouping pixels into uniform superpixels (SLIC) to extract high-level feature vectors instead of raw pixels.

Method: Training a Random Forest classifier using a 9-dimensional vector (Color, Texture, Geometry) for regional classification.



SYSTEM WORKFLOW (PIPELINE)



FEATURE EXTRACTION

This is what makes this model superior to the standard Random Forest algorithm:

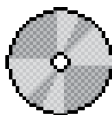
Feature Type	Method	Main Function
BGR x 3	BGR (Blue, Green, Red)	raw color data captured directly by the camera
HSV x 3	HSV (Hue, Saturation, Value)	Identifying the characteristics of the dark color of the hole.
Texture	Local Binary Pattern (LBP)	Distinguishes between asphalt roughness and pothole depth.
Tepi/Edge	Sobel Gradient	Detecting sharp edges at the edges of potholes.
Cekungan/ Depression	Blackhat Morphology	Highlighting small dark features (potholes) in bright areas.

FEATURE EXTRACTION



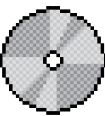
Resizing

the image is resized using cv2.resize() to speed up computation.



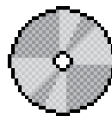
Pixel Normalization

all pixel values are divided by 255 so the range becomes [0, 1] before further processing.



Color Space Conversion

BGR is converted to Grayscale (for gradient & blackhat computation) and to HSV (for color features).



Generate On CSV with Label

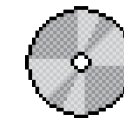
=== 5 Superpixel LUBANG (label=1) dari Val Set ===											
image_name	superpixel_id	BGR Blue	BGR Green	BGR Red	HSV Hue	HSV Sat	HSV Val	Gradient Mag	Blackhat	LBP	label
train_004	157	0.25890	0.25458	0.26893	0.55036	0.07592	0.27068	0.12250	0.50471	0.58252	1
train_004	158	0.28130	0.27553	0.29256	0.60267	0.07806	0.29491	0.13686	0.47386	0.57702	1
train_004	161	0.19203	0.18989	0.20094	0.49514	0.10591	0.20461	0.14729	0.64471	0.60534	1
train_004	162	0.22491	0.21751	0.22793	0.65668	0.07205	0.23207	0.11195	0.53280	0.61735	1
train_004	165	0.29279	0.28877	0.30678	0.82425	0.06544	0.30691	0.10969	0.40439	0.57288	1
=== 5 Superpixel JALAN (label=0) dari Val Set ===											
image_name	superpixel_id	BGR Blue	BGR Green	BGR Red	HSV Hue	HSV Sat	HSV Val	Gradient Mag	Blackhat	LBP	label
train_004	0	0.64448	0.58515	0.57205	0.64153	0.11266	0.64448	0.06300	0.09384	0.49198	0
train_004	1	0.58904	0.53754	0.52968	0.65062	0.10190	0.58904	0.03809	0.09096	0.51688	0
train_004	2	0.58463	0.53121	0.52265	0.64967	0.10677	0.58463	0.03088	0.08577	0.56107	0
train_004	3	0.56499	0.51580	0.50787	0.65004	0.10167	0.56499	0.01546	0.06941	0.55420	0
train_004	4	0.58186	0.52735	0.51991	0.65290	0.10745	0.58186	0.01448	0.03387	0.59807	0

SUPERPIXEL AGGREGATION & LABELING



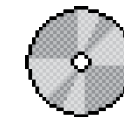
Classifying individual pixels is computationally heavy and ignores the context of the surroundings. So we use SLIC Segmentation to group pixels into Superpixels.

Segmentation



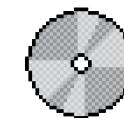
The SLIC algorithm creates compact regions based on the color and proximity

Feature Aggregation



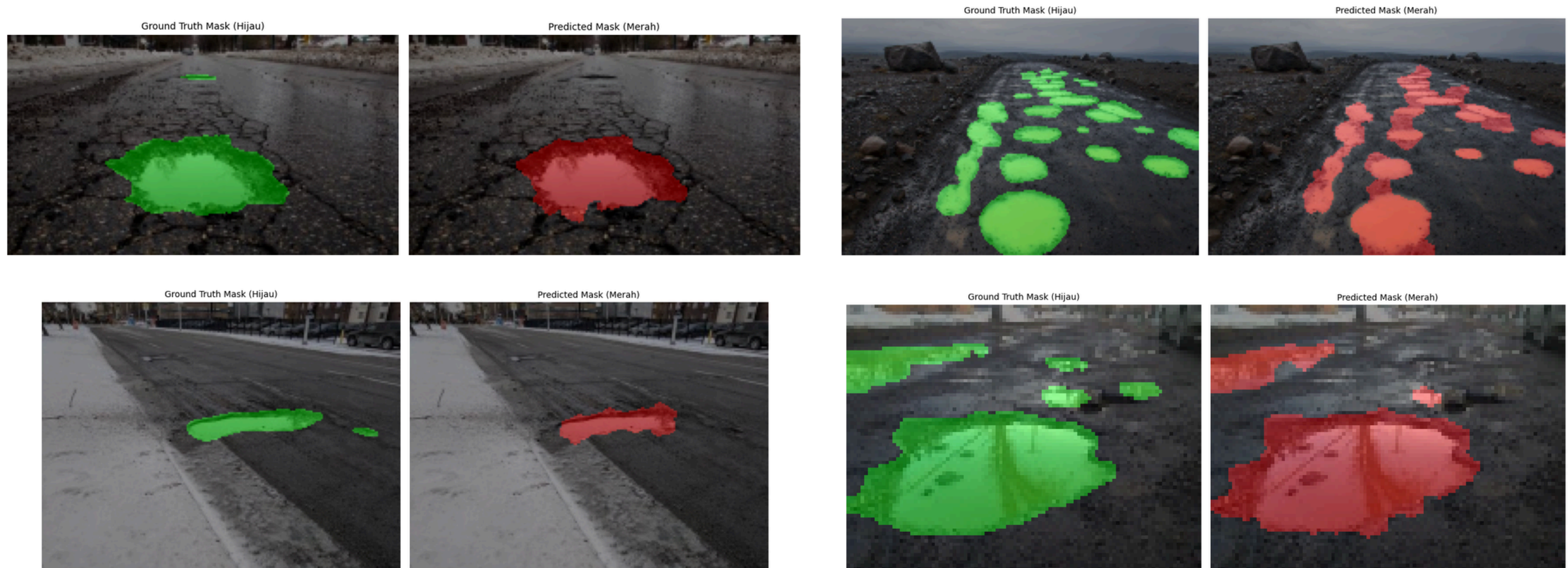
For every region in the image, we take the mean of all the pixel features (Color, Sobel, Blackhat, LBP) inside that region.

Threshold Labeling



A label is assigned to each superpixel based on the ground truth, if at least 40% of the region overlaps with a known pothole, it is classified as a pothole

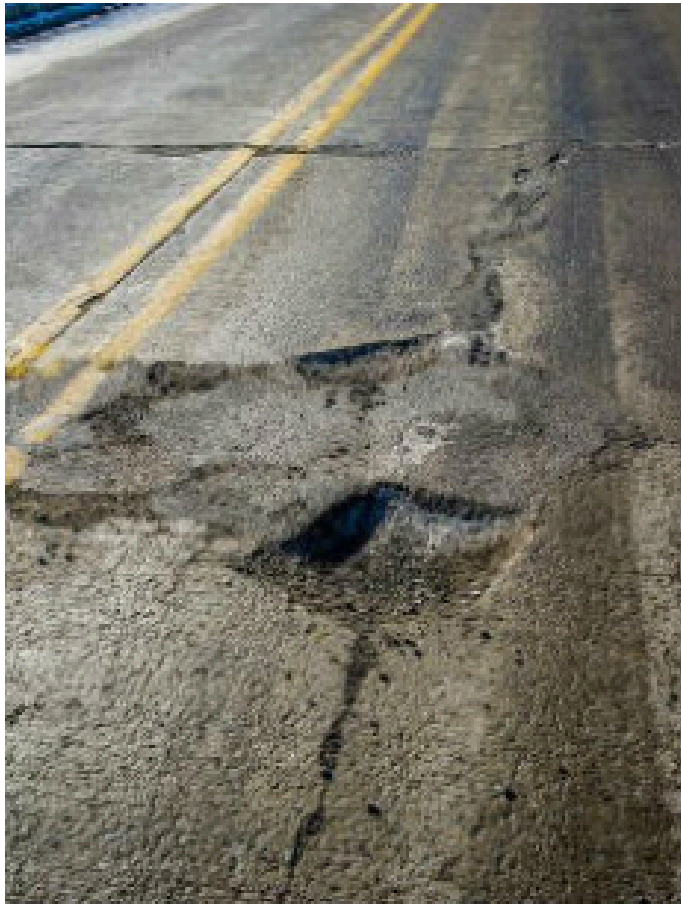
DEMO MODEL: DETECTION VISUALIZATION



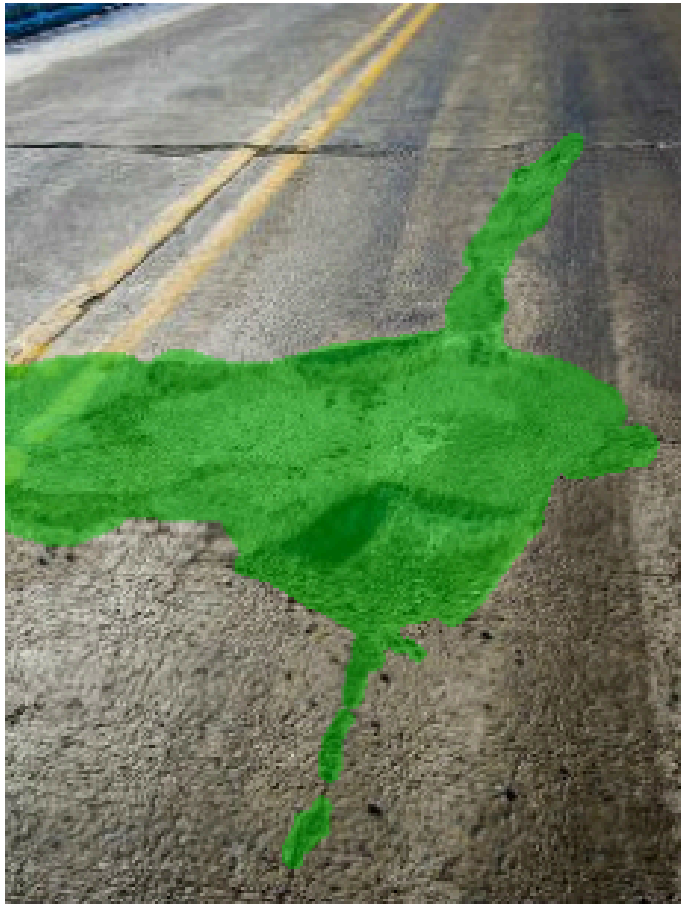
The red color indicates areas identified by the Random Forest as high-probability holes.

METHOD COMPARISON

Original



Ground Truth



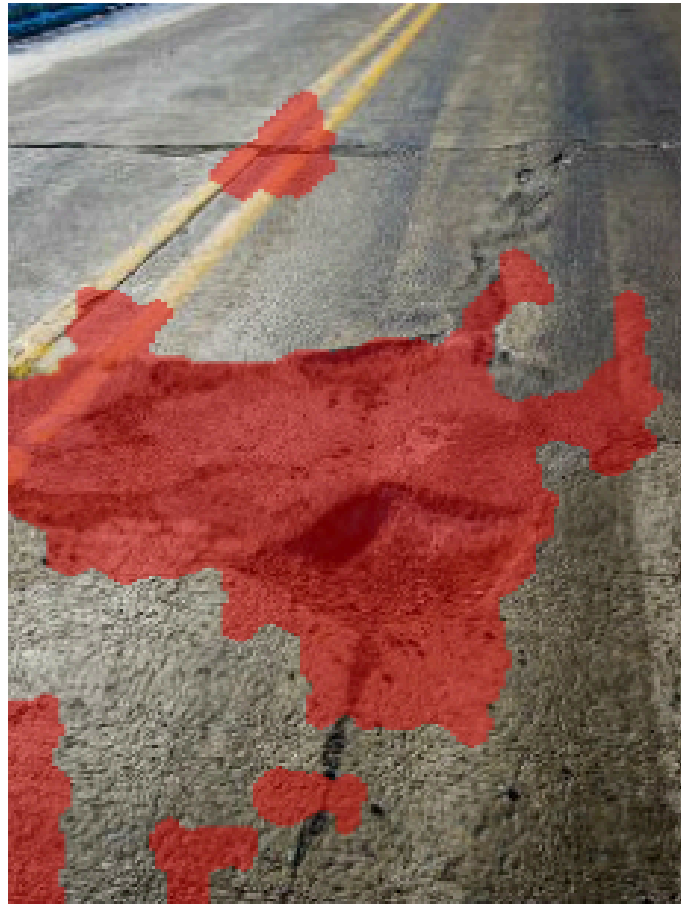
Baseline (AT)
IoU=0.309 Dice=0.472



KMeans
IoU=0.257 Dice=0.409



Random Forest
IoU=0.572 Dice=0.728



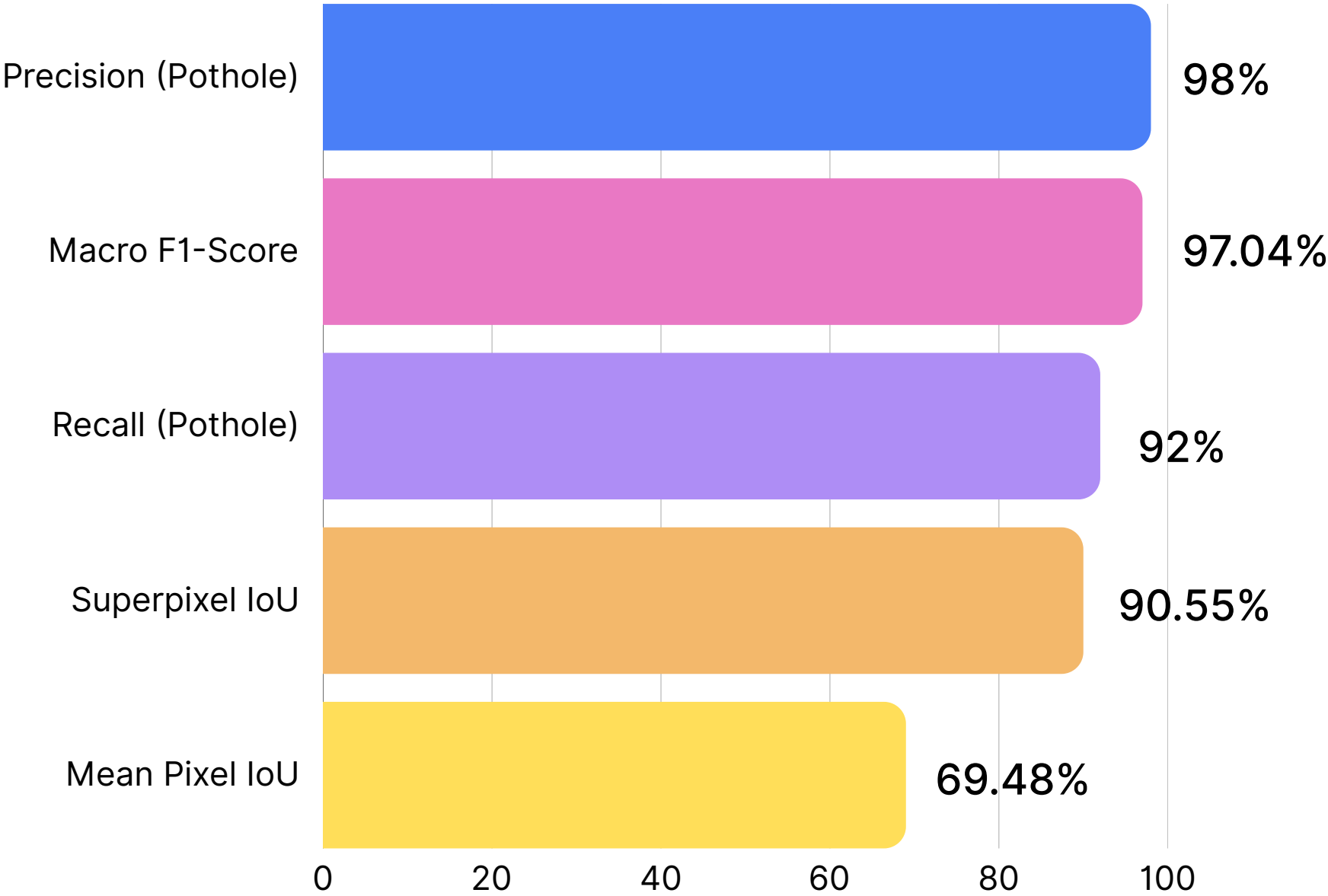
tested using external image

LETS TRY ON APPS...

MODEL EVALUATION RESULTS

98.40%

Overall Accuracy



“The model successfully exceeded the initial target of mIoU > 0.6, achieving a result of 0.69 at the pixel level.”

ADVANTAGES OF THE FINAL MODEL

01

Robustness

The LBP and Sobel features help the model ignore shadows, which were previously a major problem in K-Means.

02

Efficiency

By using only 100 trees in the Random Forest, the model is very lightweight and can be run on Edge Devices.

03

Interpretability

We can see which features are most important in identifying potholes.

FUTURE WORKS

01

Real-Time Integration

Deploying the model into dashboard cameras or IoT-based camera devices for live, real-time pothole detection.

02

Environmental Adaptability

Expanding the dataset to test the model's reliability under certain conditions, such as night time and heavy rain.

03

Automated Reporting

Developing an automated pipeline that connects the detection output with GPS mapping APIs to instantly log pothole locations for public use.



THANK YOU!

"Smart Vision for Safer Roads"

